



Outcomes-based course syllabus

COURSE INFORMATION						
Course Code	COMP001	Course Title	INTRODUCTION TO COMPUTING		Course Credit	3 Units
Course Description		This course is designed to provide students with an overview of computing fundamentals. The materials covered in this course include concepts in computers such as elements of the computer system, number system operation and conversion, data representation, digital logic system, levels of programming, computer networks, computer applications, current trends and issues. The topics prepare the students for their next subjects in the curriculum and broadens their perspective on how information technology affects every aspect of human life.				
Pre-Requisites		None		Co-Requisites	N/A	
VISION		PUP: The National Polytechnic University (PUP: Pambansang Politeknikong Unibersidad)				
MISSION		Ensuring inclusive and equitable quality education and promoting lifelong learning opportunities through a re-engineered polytechnic university by committing to: - provide democratized access to educational opportunities for the holistic development of individuals with global perspective - offer industry-oriented curricula that produce highly-skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building - embed a culture of research and innovation - continuously develop faculty and employees with the highest level of professionalism - engage public and private institutions and other stakeholders for the attainment of social development goal establish a strong presence and impact in the international academic community				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of university services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



INSTITUTIONAL LEARNING OUTCOMES (ILO)	1. Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. 2. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. 3. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry and other sectors. 4. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state-of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. 5. Passion for Lifelong Learning. Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning. 6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS	The college goals are as follows: 1. Quality IT education relevant and responsive to the changing needs of the industry and society; 2. Strong linkages with industry and other research and development institutions. 3. Strong outreach and extension programs. 4. Highly employable graduates. 5. Innovative systems and researches with greater reliance on technology. 6. Highest degree of commitment and professionalism among administrators, faculty, staff and students.



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PROGRAM LEARNING OUTCOMES (PLO)		Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	1)Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.	✓			✓					
	2)Identify, analyze, formulate, research literature, and solve complex computing problems and requirements reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.	✓			✓					
	3)Apply knowledge and understanding of information security issues in relation to the design, development and use of information systems.	✓			✓					
	4)Design and evaluate solutions for complex computing problems, its systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✓		✓	✓					✓
	5)Perform effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings.		✓	✓			✓		✓	
	6)Communicate effectively with the computing community and with society-at- large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.		✓							
	7)Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and be guided by the adoption of appropriate, ethical and legal practices.			✓				✓		✓



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	8)Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.				✓	✓			✓	
PERFORMANCE INDICATORS (PI)		Alignment to PLOs								
		1	2	3	4	5	6	7	8	
	• Simplify equations using various methods for logic simplifications. • Design, implement, test and debug a program, based on a given specification	✓								
	• Complete and successfully defend a thesis in line with the discipline. • Design and develop a computing solution using object-oriented approach • Implement solutions to circuit design problems using various logical gate functions. • Design a database based on user requirements using a widely used modeling notation		✓							
	• Design an overall architecture of the system and justify its appropriateness • Identify the strengths and weaknesses associated with different approaches to security. • Describe the set of controls and processes intended to protect information and information systems			✓						
	• Develop appropriate user interfaces for domain specific applications. • Evaluate how system and infrastructure components are organized into solutions in different environments.				✓					



	• Discuss with team members different software methodologies and techniques and identify best solutions for implementation. • Present a proposed solution in class or in a public forum • Justify the proposed solutions effectively through the use of scientific methodologies and document the results properly. • Analyze solutions employed by organizations to address different computing issues.					✓			
	• Immerse/expose in an actual working environment in industry. • Propose an ethical and feasible software solution to the identified research problem that employs new designs, tools, and methodologies. • Evaluate professional, ethical and social issues of computing decisions.							✓	
	• Create a report on a conducted independent learning activity. • Identify the relevant/legitimate sources of technology based information and utilize them for continuous learning								✓
COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	• Explain fundamental principles, concepts and evolution of computing systems as they relate to different fields.			✓	✓		✓		
	• Perform number conversion, data representations and relate number systems to foundational knowledge on digital logic systems		✓						



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	Expound on the recent developments in the different computing knowledge areas						✓		✓
	Analyze solutions employed by organizations to address different computing issues	✓		✓	✓				
	Explain how information technology impacts important social issues, what social values it contributes, and its impact to human life in general					✓		✓	
	Demonstrate awareness of the important social and ethical issues related to information technology							✓	

OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design			Assessment	Alignment to CLOs					
			Face-to-Face	Flexible Learning and Teaching Activities (FLTAs)			1	2	3	4	5	6
				Synchronous	Asynchronous							
1	a. Demonstrate an understanding of what the subject is all about, what will be in scope for the semester, and what students are expected to learn b. Communicate with fellow students and teacher and begin to establish rapport c. Identify and explain the course assessment and validation criteria , including grading	1. Introduction to the Course a. Vision Mission Goals and Objective of the University, and College. b. Self-Introduction c. Course Overview d. Grading System e. Classroom Management	Discussion of the different learning content , grading system, and rules inside the f2f class Introductions	Discussion of the different learning content, grading system, and rules inside the online class Introductions Via zoom meeting								



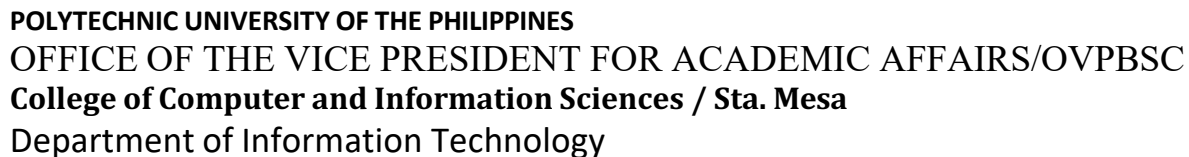
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	system to understand how to pass the subject a. Explain what are the do's and don'ts while the class is on-going											
2	a. Categorize computers. b. Contrast elements of computer system. c. Identifies various events/improvements in the computing world. d. Qualifies the understanding of computer usage	Unit 1: Overview of Information and Communications Technology 1. Introduction to Computers 2. Elements of a Computer System 3. Classification of Computers 4. Capabilities and Limitations of Computers 5. History of Computing	Lecture and discussions Video presentation Use of tools - mentimeter, mural, jamboard	Lecture and discussions Video presentation Use of tools - mentimeter, mural, jamboard	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video.	Examination	✓					
3-8	a. Distinguish the various number system and data representation b. Perform number system operation such addition, subtraction and complementation. c. Transform numeric and non-numeric data to how they are	Unit 2: Number System 1. Number System Operation and Conversion (Whole Number and Fractional Conversion) 2. Number Systems Arithmetic (Addition and Subtraction) 3. Data Representation	Lecture and exercises Recitation Assignment Exercises	Lecture and exercises Recitation via the chat box Assignment Exercises	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Examination Graded Exercises on number system conversion, arithmetic, and data representation (homework)	✓					



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	stored inside the computer system													
9	MIDTERM													
10-11	Part 1 a. Explain the functions of the main units of a physical computer system b. Describe what input and output devices are c. Explain the workflow of the processor/memory/i nput and output devices and architecture d. Differentiate the primary and secondary storage devices Part 2 e. Define what boolean algebra is f. Identify the different logic gates	Unit 3: Hardware Part 1 : Components of the system unit 1. Processor 2. Memory 3. Input and Output Devices Part 2 : Digital Logic System	Lecture and exercises Recitation Exercises on simple circuits	Lecture and exercises Recitation via the chat box Exercises on simple circuits	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Examination Graded Exercises on creating simple circuits and its equivalent boolean equation and truth tables (homework)	✓	✓						



	g. Illustrate the representation of the different logic gates h. Convert boolean algebra expression into a logic circuit i. Create truth tables for the corresponding logic circuits and boolean expression j. Explain basic theorems and postulates on digital logic system											
12	a. Identify different jobs available for an ICT professional. b. Contrast roles and responsibilities for the different jobs in the ICT profession.	Unit 4: Peopleware 1. Jobs in IT Industry 2. Roles and responsibilities of different jobs in IT	Lecture and Discussion Reporting Recitation	Lecture and Discussion Reporting Recitation via the chat box	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Examination Graded Report			✓	✓		
13	a. Differentiate an application software from a system software b. Explain the types of a system software c. Discuss the different means by which application softwares are made available	Unit 5: Software 1. System Software a. Operating System b. Utility Programs c. Compilers and Interpreters 2. Application Software 3. Software distribution methods	Lecture and Discussion Recitation	Lecture and Discussion Recitation via the chat box	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Examination	✓					



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14	a. Discuss network models b. Explain network topologies c. Explain internet concepts	Unit 6: Computer Networks 1. Concepts in Computer Networking 2. Network Services 3. Internet and the World Wide Web	Lecture and Discussion Recitation	Lecture and Discussion Recitation via the chat box	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Examination									
15	a. Demonstrate awareness on the current ICT trends and social issues. b. Relate the current ICT trends and social issues on studies and their impact on society	Unit 7: Computer Trends and Issues in ICT 1. Current Trends 2. Social and Legal Issues	Lecture and Discussion Recitation	Lecture and Discussion Recitation via the chat box	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Examination				✓	✓	✓	✓		
16-17	a. Explain the difference between AI, machine learning, and deep learning b. Provide applications of AI in different	Unit 8: Special Interest Topics in ICT 1. Artificial Intelligence 2. Data Science and Big Data 3. Internet of Things	Lecture and Discussion Recitation	Lecture and Discussion Recitation via the chat box	Watch video of lecture Submit answers to questions provided Research on additional information on the topic discussed in the video	Examination				✓	✓	✓	✓		



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	industries and in daily use. c. Explain supervised, unsupervised learning and other concepts related to AI d. Explain what the field of data science is e. Identify the skills/expertise needed to be a data scientist f. Discuss what big data is and how it relates to data science g. Explain what IoT is. h. Identify different objects attached to the internet																	
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18	FINAL EXAMINATIONS
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LIST OF REFERENCES

1. Dale, Nell - Computer science illuminated, copyright 2020
2. Assad Abbas, Samee U. Khan, and Albert Y. Zomaya - Fog Computing: Theory and Practice, copyright 2020
3. I. Foster, G. R. Joubert, and L. Kučera - Parallel Computing: Technology Trends, copyright 2020
4. Jianwei Huang, Yang Yang, Tao Zhang, and Joe Weinman - Fog and Fogonomics : Challenges and Practices of Fog Computing, Communication, Networking, Strategy, and Economics, copyright 2020
5. David Evans - Introduction to Computing Explorations in Language, logic, and Machines (eBook), copyright 2019



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Note: Extended Readings may be assigned by the professor.

CLASSROOM POLICY (to be filled out by the assigned faculty)

FACE-TO-FACE DELIVERY

1. The course is expected to have a minimum of four (4) quizzes.
2. Assignments and projects/report works will be given throughout the semester. Such requirements shall be due as announced in class.
3. Students are required to attend classes regularly, including possible make-up classes. The student will be held liable for all topics covered and assignments made during his/her absence. The university guidelines on attendance and tardiness will be implemented.
4. Any evidence of copying or cheating during any examinations may result in a failing grade from the examination for all parties involved. Note that other university guidelines shall be used in dealing with this matter.
5. Students are advised to keep graded work until the semester has ended.
6. Contents of the syllabus are subject to modification with notification.
7. Cell phones, radios or other listening devices are not allowed to be used inside lecture and laboratory rooms to prevent any distractive interruption of the class activity. *
8. No foods, drinks, cigarettes nor children are allowed inside the lecture and laboratory rooms. *
9. Withdrawal and dropping from the subject should be done in accordance with existing university policies and guidelines regarding the matter.

FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)

1. Attendance and Tardiness
 - a. Classes start promptly on designated/agreed time
 - b. Attendance will be checked regularly
2. Internet Connection
 - a. Internet connection may be interrupted from time to time. Students are encouraged to go right back to the meeting as soon as the connection is restored
3. All materials including video recording will be uploaded to our LMS
4. During lectures
 - a. Short discussion will follow at end of each leacture to check student understanding of the topic.
 - b. Students will be asked to read from the presentation from time to time
 - c. Students are encouraged to take down notes from the presentation
5. GAD/PWD Considerations
 - a. All, regardless of gender will be treated equitably, with respect
 - b. Proper considerations will be accorded to those with physical limitations

(Read also netiquette guidelines for online students)



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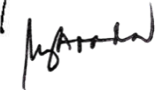
COURSE REQUIREMENT/S	GRADING SYSTEM									
	First Half = 70% (Class Standing) + 30% (Midterm Exam) Second half = 70% (Class Standing) + 30% (Final Exam) Class Standing refers to quizzes, homeworks, reports First Half Grade + Second half Grade = Final Grade									
CLASS INFORMATION	FACULTY INFORMATION									
Section:	Name of Faculty:									
Time:	Consultation Time:									
Room:	Office Tel. No./ Mobile Phone No.									
Semester:	Institutional Email:									
Suggested Rubrics for Assessment (to be filled out by the assigned faculty)										
Suggested rubrics for assessment: A. Essay Rubrics B. Portfolio Rubrics C. Students shall be graded or marked in accordance with the following system based on the Student Handbook:										
<table border="1"><thead><tr><th>GRADE/MARK</th><th>PERCENTAGE/ EQUIVALENCE</th><th>DESCRIPTION</th></tr></thead><tbody><tr><td>1.0</td><td>97-100</td><td>Excellent</td></tr><tr><td>1.25</td><td>94-96</td><td>Excellent</td></tr></tbody></table>		GRADE/MARK	PERCENTAGE/ EQUIVALENCE	DESCRIPTION	1.0	97-100	Excellent	1.25	94-96	Excellent
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1.50	91-93	Very Good
1.75	88-90	Very Good
2.0	85-87	Good
2.25	82-84	Good
2.50	79-81	Satisfactory
2.75	76-78	Satisfactory
3.0	75	Passing
5.0	65-74	Failure
Inc.		Incomplete
W		Withdrawn

This is for Main Campus use.

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
<u>Faculty/Committee/Date</u>	 MELVIN C. ROXAS  MARIAN G. ARADA <u>Chairperson/Date</u> <u>Chairperson/Date</u>	 BENILDA ELEONOR V. COMENDADOR <u>Dean/Date</u>	 EMANUEL C. DE GUZMAN <u>Vice President for Academic Affairs/Date</u>

This is for the Branches and Campuses use.



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Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
<u>Faculty/Committee/Date</u>	<u>Head, Academic Program/Date</u>	<u>Director/Date</u>	<u>Vice President for Branches & Satellite Campuses/Date</u>

Form No. _____
Revision No. _____
Date of Approval _____
Semester _____
Academic Year _____